iLearn Practice Exam

10BR · Chapter / Topic 6

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 6: Automated and Emerging Technologies

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 75

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 12 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 A company uses robots in its factory to manufacture large pieces of furniture.

(a) One characteristic of a robot is that it is programmable.

State **two** other characteristics of a robot.

1 .....

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2 .....

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[2]

(b) Give **two** advantages to company employees of using robots to manufacture large pieces of furniture.

1 .....

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2 .....

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[2]

(c) Give **one** disadvantage to the company's owners of using robots to manufacture large pieces of furniture.

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..... [1]

2 One component of an expert system is the inference engine.

(a) Identify the **three** other components in an expert system.

1 .....

2 .....

3 .....

[3]

(b) Describe the role of the inference engine in an expert system.

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..... [2]

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3 A business has a system that is described as having artificial intelligence (AI).

(a) State **one** of the main characteristics of an AI system.

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..... [1]

(b) An AI system is an expert system.

Explain how an expert system operates.

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..... [6]

4 A car repair garage uses an expert system.

(a) Complete the description about the operation of the expert system.

Use the terms from the list. Some of the terms in the list will **not** be used.

inference engine interface knowledge base

machine learning mechanical engine output device

question base rule base

An expert system has a ..... that contains a list of facts. The ..... applies the ..... to the ..... to reach a diagnosis for the repair of the car. The user provides data to the system using an ..... . [5]

(b) The expert system has machine learning capabilities.

Describe what is meant by machine learning capabilities.

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..... [4]

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5 A garage uses an expert system to help diagnose any problems with cars that need repair.

(a) The expert system is an example of artificial intelligence (AI).

Describe what is meant by AI.

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..... [2]

(b) A car has a problem with its braking system, so it is brought into the garage.

Explain how the expert system operates and how it is used to help diagnose the problem.

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..... [5]

6 A doctor's surgery has an expert system that helps diagnose the illnesses of its patients.

(a) The expert system has the ability to automatically adapt its own processes and data.

Give the name of this ability.

..... [1]

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7 A robot vacuum cleaner uses sensors to navigate around obstacles in a room.

(a) An expert system can be used to diagnose an error with a robot vacuum cleaner.

Circle **three** components that are part of an expert system.

knowledge base operating system firewall

server printer actuator inference engine

rule base encryption

[3]

(b) Explain how the robot vacuum cleaner can make use of machine learning.

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..... [3]

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8 A search engine uses artificial intelligence (AI) to identify the most relevant web pages for the user.

(a) One characteristic of AI is the ability to learn.

Identify **three** other characteristics of AI.

1 .....

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2 .....

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3 .....

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[3]

(b) The search engine uses machine learning.

Explain how machine learning is used by the search engine.

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..... [3]

9 A robot is used to move car parts in a factory.

(a) One characteristic of the robot is that it has a mechanical structure.

Give **one** other characteristic of a robot.

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..... [1]

(b) The robot has firmware.

State what is meant by firmware.

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..... [1]

(c) Give **two** advantages to the workers in the factory of using robots to move the car parts.

1 .....

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2 .....

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[2]

(d) The robots have the ability to automatically adapt their own processes.

State the name of the ability described.

..... [1]

10 A computer game is programmed to use artificial intelligence (AI) to create the movements and actions of the computer-controlled characters in the game.

(a) Describe what is meant by AI.

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..... [3]

(b) Explain how the AI can be used to improve the movements and actions of the computer-controlled characters.

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..... [3]

11 A child has a robot animal. The robot animal can perform actions like making a noise and walking around a room.

If the robot animal hears a clapping sound, it makes a noise.

If the robot animal is walking and it senses an object is less than 10 cm away, it will stop.

(a) Give **one** suitable sensor that the robot animal could use to identify a clapping sound.

..... [1]

(b) Give **two** suitable sensors that the robot animal could use to identify if an object is nearby.

1 .....

2 .....

[2]

(c) One characteristic of the robot animal that makes it a robot is that it has electrical components that include sensors, a microprocessor and actuators.

(i) Give **one** other characteristic of the robot animal that makes it a robot.

..... [1]

(ii) The sensor continuously sends digitised data to the microprocessor.

Describe how the microprocessor uses the data from the sensor to stop the robot animal when an object is less than 10 cm away.

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..... [3]

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(d) The robot animal uses artificial intelligence (AI).

Explain how the robot animal can use AI to learn where objects are in a room.

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..... [4]

12 A computer game uses artificial intelligence (AI) to create extra characters (players) in the game.

(a) One characteristic of the AI is that it can adapt and learn.

Give **one** other characteristic of the AI.

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..... [1]

(b) The AI uses machine learning.

State what is meant by machine learning.

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..... [1]

(c) Explain **one** example of how the AI could use machine learning to create extra characters.

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..... [2]

(d) The computer game has an expert system that can be used to find a solution to any problems with the game. The expert system uses an inference engine.

Explain the role of the inference engine in finding a solution to a problem with the game.

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..... [3]

END OF QUESTION PAPER